

SCRATCH STARTER LAB

Make Your First Animation

Meet Pixel and learn how to code
with colorful puzzle blocks!

LEA LABS • SCRATCH FOR KIDS



What can Scratch do?

Interactive Stories

Make your own characters talk, move, and go on exciting adventures!

Fun Games

Build games you can actually play, with scores, levels, and winning rules.

Cool Animations

Bring your art to life with motion, sounds, and special effects.

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The Stage & Sprites

THE STAGE

The screen where your game or story plays out.

SPRITES

Characters and objects in your project.

BACKDROPS

The picture behind everything.



The Code Blocks Playground

COLOR-CODED

Every color does a different job! Blue is for moving, purple is for looks, and pink is for sound.

SNAP TOGETHER

Blocks connect just like puzzle pieces. If they fit together, they work together!

MAKE A SCRIPT

When you stack blocks together, it's called a "script". It tells your sprite exactly what to do.

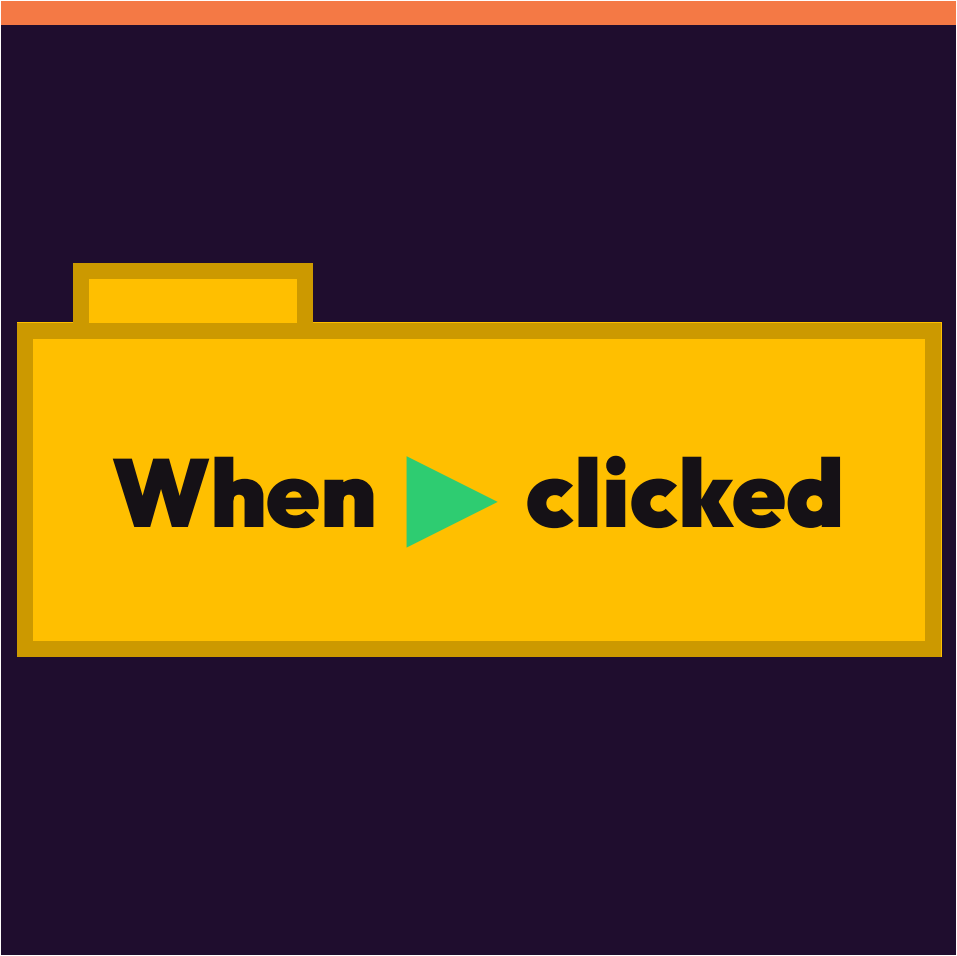


Events: The Green Flag

Every great story needs a beginning. In Scratch, we use **Event Blocks** to tell our code exactly when to wake up and start running.

THE GREEN FLAG

The most important event block is the Green Flag. When you click the Green Flag button above the stage, all scripts starting with this yellow block will begin at the exact same time!

A large yellow Scratch event block with a green flag icon and the text 'When clicked' is shown. It is positioned in front of a dark blue background that represents a Scratch script area. The block has a yellow border and a small yellow tab on its top-left corner.

When  **clicked**

Motion: Make a Move

BLUE BLOCKS

Motion blocks are always blue. They control where your sprite goes!

MOVE & TURN

Use "move 10 steps" to walk forward, or "turn 15 degrees" to spin around.

TELEPORT

Use "go to" blocks to instantly jump to a specific spot on the stage.



Looks: Costumes and Say

Purple blocks control how your sprites look and talk!

CHANGE COSTUMES

A "costume" is a different picture for the same sprite. By switching costumes quickly, you can make a character look like it's walking or dancing!

next costume

SAY HELLO!

Want your character to speak? The "say" blocks make a comic-book speech bubble appear right above your sprite.

say [Hello!] for (2) seconds

Sound: Add Music & Noise

Make your project loud and fun with the pink Sound blocks!

SOUND EFFECTS

Play a quick noise like a "Meow", a "Pop", or a laser blast when something exciting happens.

BACKGROUND MUSIC

Add a song that plays the whole time your game or story is running.



Timing: Wait and Repeat

SLOW DOWN!

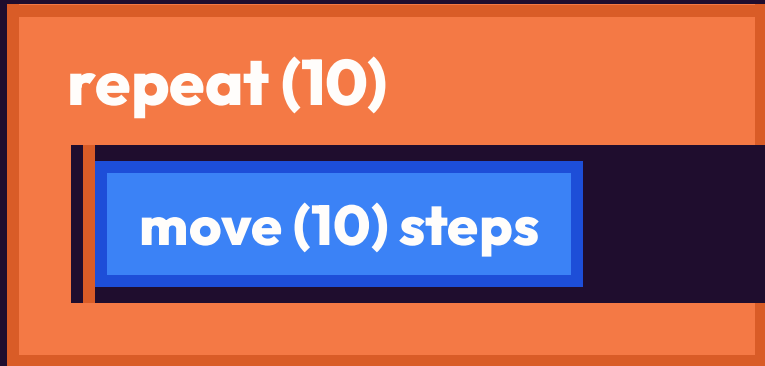
The orange **Wait** block makes your script pause before moving to the next block. It's perfect for timing conversations or slowing down an animation.

An orange Scratch 'wait' block with the text 'wait (1) seconds' in white. The block has a 3D effect with a darker orange shadow.

wait (1) seconds

DO IT AGAIN!

The **Repeat** block is a loop. It wraps around other blocks like a hungry mouth and runs them over and over again!

A Scratch 'repeat' block. It is an orange block with a 'repeat (10)' label. Inside the block, there is a blue 'move (10) steps' block. The repeat block has a 3D effect with a darker orange shadow.

repeat (10)

A blue Scratch 'move' block with the text 'move (10) steps' in white. The block has a 3D effect with a darker blue shadow.

move (10) steps

Build-Along: Cross the Stage

STEP 1: START

Drag a yellow **"When Green Flag clicked"** block to start your script.

STEP 2: POSITION

Snap a blue **"go to x: -200 y: 0"** block so your sprite always starts on the left side.

STEP 3: LOOP

Add an orange **"repeat 10"** block. This will make our animation happen over and over!

STEP 4: MOVE & WAIT

Inside the loop, put a blue **"move 10 steps"** and an orange **"wait 0.1 seconds"**. Click the flag and watch them go!



Debugging: When Things Get Stuck

WHAT IS A BUG?

A bug is a mistake in your code. It's normal—every coder fixes them step by step.

BE A DETECTIVE

Read your blocks top to bottom. Check colors and that blocks are snapped together.

TEST AND FIX

Change one block, click the Green Flag, and observe. Try again if needed.



Lab Launch: Animated Story

— YOUR MISSION

- Pick a cool backdrop and at least 2 sprites.
- Use the Green Flag to start your story.
- Make your characters move and talk.
- Add a fun sound effect!

★ BONUS CHALLENGE

Can you use an orange "repeat" block to make one of your characters do a silly dance?

